

ECE501: Digital Image Processing

Project Topic: Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

GROUP 19

Week 4 Progress Report

Date: October 10, 2026

Deliverable: Implementation and comparison of LBP and GLCM defect detection on the same defect-free and defective fabric images

Group Members:

Name	Enrolment No.
Pal Joshi	AU2440172
Nishka Shah	AU2440168
Mahee Shah	AU2440174
Jashvi Shah	AU2440026

Objectives of the Week

- Implement the LBP-based [1] and GLCM-based [2] detection methods studied in Weeks 2 and 3.
- Run both methods on the same defect-free and defective images.
- Compare the outputs in terms of detection, false alarms, localization and run time.

Work Done This Week

A. Dataset and setup

- Dataset: AITEX Fabric Image Database (AFID) [3], which provides pixel-level ground-truth masks; fabric type 03 (defect-free folder 2311694-1930c7u). Images are 4096 x 256 pixels and are processed whole, without resizing.
- Images used: 8 defect-free images as the normal reference, 5 other defect-free images to count false alarms, and 5 defective images with ground-truth masks (at most 15).
- Common pre-processing for both methods: grayscale conversion and percentile contrast stretching (2nd to 98th percentile).

- Both methods were run on the same images, with the same 32 x 32 patches and stride 16.

B. LBP method

- Each patch is described by a 10-bin normalised histogram of rotation-invariant uniform LBP codes ($P = 8$, $R = 1$) [4].
- The normal reference is the set of histograms of all patches from the 5 defect-free reference images.
- The anomaly score of a patch is its chi-square distance to the nearest defect-free patch. Scores of overlapping patches are averaged into a score map.
- The threshold is a percentile (95, 99 or 99.9) of leave-one-image-out scores of the reference images. Paper [1] instead uses a log-likelihood ratio and the maximum score of defect-free windows.
- Pixels above the threshold are marked defective. Regions smaller than 32 pixels are removed and the mask is smoothed by morphological closing.

C. GLCM method

- Each patch is described by five GLCM features: contrast, dissimilarity, homogeneity, energy and correlation, at distances 1 and 2 and angles 0, 45, 90 and 135 degrees (averaged), with 16 gray levels.
- Features are standardised using the mean and standard deviation of the reference patches.
- The anomaly score is the RMS distance to the nearest defect-free patch. The same percentile threshold, score-map averaging and post-processing are used as for LBP.

D. Comparison conditions

Both methods follow the same pipeline, and only the feature extraction and scoring steps differ:

1. Pre-process the test image once, so both methods start from the identical image.
2. Divide it into identical patches for both methods.
3. Extract LBP features and calculate LBP-based anomaly scores.
4. Extract GLCM features and calculate GLCM-based anomaly scores.
5. Apply each method's own threshold and generate its predicted defect mask.
6. Display the original image, LBP output and GLCM output side by side.
7. Compare detection metrics (Accuracy, Precision, Recall, F1) using labelled test images, if available.

The same test images, pre-processing and window grid were used for both methods, so that differences come from the descriptor and not from the setup.

E. Changes from the planned design (Weeks 2 and 3)

While implementing, we simplified some choices from the earlier design so that both methods could run on the same footing:

- Patch size: 32 x 32 with stride 16, instead of 128 x 128, because the defects in this dataset are thin and small.
- Normal reference: all reference-patch features are kept and each test patch is scored by its distance to the nearest defect-free patch, instead of a single mean histogram (LBP) or a mean-and-covariance model with Mahalanobis distance (GLCM). This avoids inverting a covariance matrix estimated from few samples, the difficulty noted in Week 3.
- Threshold: a percentile of leave-one-image-out scores of the reference images, instead of mean + k times standard deviation.
- Pre-processing: percentile contrast stretching only. Gaussian or median denoising and CLAHE were not used in this run.
- LBP scales: only (P = 8, R = 1) is used so far; (P = 16, R = 2) and the two-scale combination are not yet included. Gabor features are planned for Week 5.

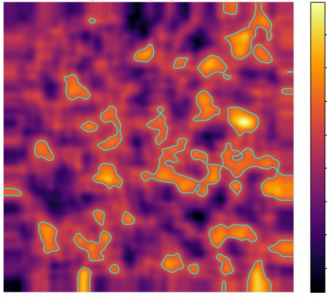
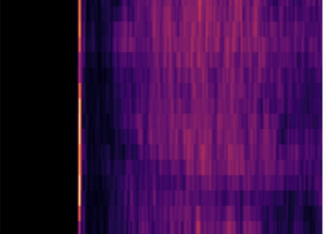
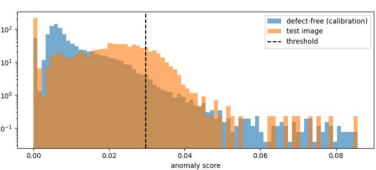
Literature Review

Topic / Paper	Key Takeaway	Relevance to Our Project
Tajeripour et al., Fabric Defect Detection Using Modified Local Binary Patterns [1]	Builds an LBP reference from defect-free fabric, scores windows by log-likelihood ratio and sets the threshold from defect-free windows. Reports detection rates above 95% on plain and patterned fabrics.	Basis of our LBP implementation.
Real Time Fabric Defect Detection System on an Embedded DSP Platform [2]	Derives co-occurrence statistics such as energy, contrast, homogeneity and correlation for real-time fabric inspection.	Basis of our GLCM implementation; shows GLCM statistics are cheap enough for fast inspection.

Results and Analysis

Table 1. Comparison of LBP and GLCM on the same image pair.

Criterion	LBP	GLCM
Defect detected?	Yes	Yes

Parameters	P = 8, R = 1 (uniform)	Distances 1 and 2, angles 0°, 45°, 90°, 135°
Anomaly Score	3. Anomaly Score Map (cyan = threshold) 	GLCM Anomaly Score Map  
Run time per image	39 sec	1 min
Threshold	0.2097	0.0298
Suspected Defective area	10.88%	14.25%

===== LBP DETECTION SUMMARY =====

```

LBP (P, R)           : (8, 1)
Patch size / stride  : (16, 4)
Threshold            : 0.2097
Defect regions found : 15
Suspected defect area: 10.88 %
Verdict              : DEFECTIVE

```

```

===== GLCM DETECTION RESULT =====
Image: 0075_010_03.png
Prediction: DEFECTIVE
Suspected defect area: 14.25 %
Anomaly threshold: 0.0298
=====

Tuning help - defect area (%) at different THRESHOLD_MARGIN values:
margin x0.5 -> 58.03 %
margin x0.75 -> 37.35 %
margin x1 -> 14.21 %
margin x1.5 -> 0.66 %
margin x2 -> 0.32 %
margin x3 -> 0.05 %

```

Analysis (write from your own observations):

- Which method found the defect more clearly, and why:
-LBP. Its yellow boxes and bright blobs are compact and localised, while GLCM gives long vertical streaks that look like the fabric's periodic pattern rather than the defect.
- Which gave more false alarms on the defect-free image:
-GLCM. It flagged 14.25% as defective vs 10.88% for LBP, and its score histograms overlap heavily.
- Which was faster or simpler to tune:
-LBP. It ran in 39 sec vs 1 min and has only two parameters (P, R), while GLCM needs distances, angles and features.
- Agreement with literature: [1] reports that LBP works on both plain and patterned fabric, while [2] uses GLCM statistics for fast, real-time inspection.
-Partly. LBP agrees with [1] since it worked on patterned fabric. GLCM did not fully agree with [2]: it was slower than LBP and less accurate here, though it was still reasonably quick.

Challenges Faced

- **Tiny defects:** a defect is much smaller than a 32×32 patch, so the score barely changes and localisation is coarse.
- **Threshold sensitivity:** a small change in the threshold swings the flagged area from 58% to almost 0%, so it is hard to pick a stable value.
- **Test image mismatch:** the defective image was brighter than the references, so both methods flagged about 70% of the image as defective.

- **False alarms:** both methods flagged 11 to 14% of the image, far more than a real defect. LBP gave scattered false spots, and GLCM reacted to the fabric margin and lighting.
- **Overlapping scores:** normal and defective patch scores overlap, so the threshold cannot separate them cleanly.

Plan for Next Week (Week 5)

- Add Gabor features and compare them with LBP and GLCM.
- Tune window size and threshold factor k .
- Compute patch-level and pixel-level metrics (Accuracy, Precision, Recall, F1, IoU, Dice) on more images.

Conclusion

In Week 4 we implemented the LBP and GLCM methods and ran both on the same defect-free and defective images. LBP performed better than GLCM because it localised the defect more clearly with fewer false alarms (10.88% vs 14.25% suspected area), and it was also faster (39 sec vs 1 min) and simpler to tune. Next week we will add Gabor features and evaluate on a larger set of images.

References

- [1] F. Tajeripour, E. Kabir, and A. Sheikhi, "Fabric defect detection using modified local binary patterns," EURASIP J. Adv. Signal Process., vol. 2008, Art. no. 783898, 2008. doi: 10.1155/2008/783898.
- [2] "Real time fabric defect detection system on an embedded DSP platform," arXiv:1410.0371. [Online]. Available: <https://arxiv.org/pdf/1410.0371>
- [3] [AITEX FABRIC IMAGE DATABASE - Aitex](#)